

WASP-35b

R-1767

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_131208 · created 2026-08-02T21:43:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456634.86762 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.033
Transit depth (Rp/Rs)^2	2.51 +/- 1.04 [%]
Orbital Inclination (inc)	87.53 +/- 1.55
Airmass coefficient 1 (a1)	0.993 +/- 0.032
Airmass coefficient 2 (a2)	0.005 +/- 0.011
Scatter in the residuals of the lightcurve fit is	3.16 %
Best Comparison Star	#2 - [444, 209]
Optimal Aperture	2.71
Optimal Annulus	6.44
Transit Duration (day)	0.13 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.158 $\pm$ 0.033 vs 0.1241 (deviation 0.9 σ)
Duration fit vs published	0.13 d vs 0.1317 d (deviation 0.13 σ)
Mid-time O–C	0.7 min
Depth SNR	4.2
Residual scatter	3.16 %

Light curve

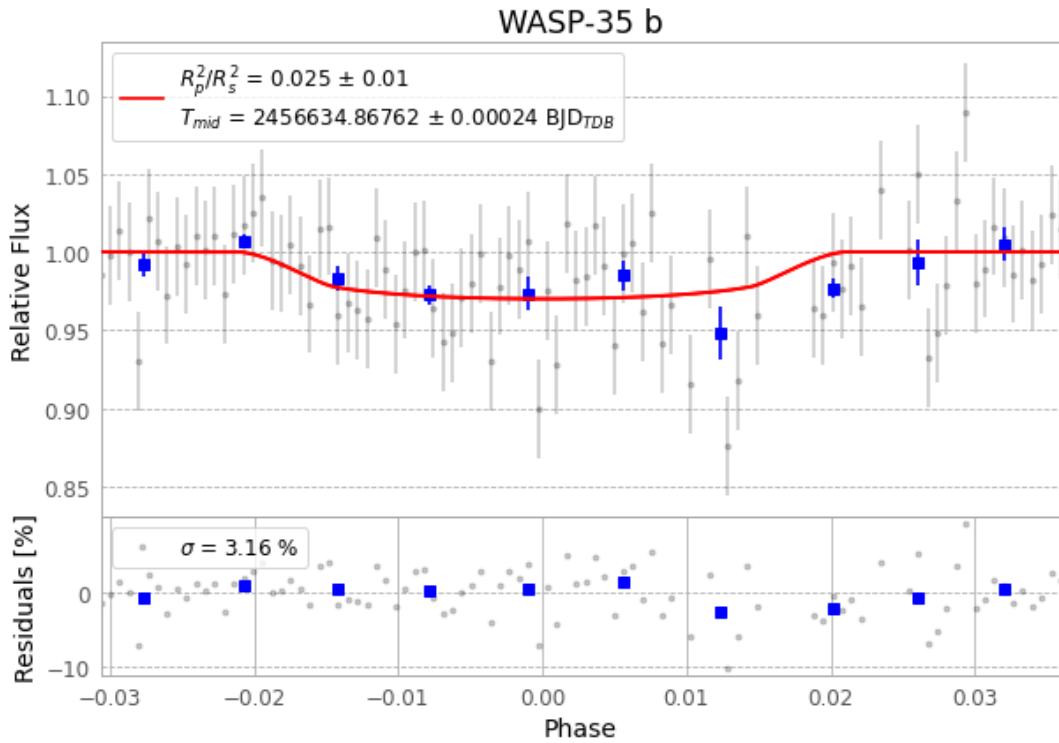


Figure 1 — FinalLightCurve_WASP-35 b_2013-12-07.png (SHA-256 09a8a975388db0fa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 39381b313602ccfe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

102 FITS frames (67 MB), WASP-35131208062112.FITS → WASP-35131208112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-131208.log (1a450eeae54e...), FinalLightCurve_WASP-35 b_2013-12-07.png (09a8a975388d...), FinalLightCurve_WASP-35 b_2013-12-07.csv (98a43806d0a1...)

Compute resources

Wall time 195.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 21:43Z.